

Retail Sales - Exploratory Data Analysis Report

Oasis Infobyte Data Analytics Internship | Level 1

1. Dataset Overview

This report presents a comprehensive analysis of 5,000 retail sales records after data cleaning. The dataset spans from 2020 to 2023 and includes transactions across three customer segments (Consumer, Corporate, Home Office), four regions (North, South, East, West), and three product categories (Furniture, Office Supplies, Technology). Key attributes analyzed include Sales, Profit, Quantity, Discount, Customer Name, State, and Order Date.

2. Methodology

The analysis followed a structured EDA workflow: (1) Data loading and initial inspection, (2) Removal of 75 duplicate records, (3) Imputation of missing values using median for numerical features and a placeholder for categorical features, (4) Feature engineering to extract temporal attributes (year, month, month name) and derived metrics (Profit Margin %, Sales per Unit), (5) Univariate, bivariate, and multivariate analysis using Pandas, Seaborn, and Matplotlib, (6) Visualization generation with professional formatting and high resolution.

3. Key Findings

Seasonal Trends: Sales peak in Q4 (holiday season) and dip in Q1-Q2. Monthly profit trends mirror sales but show sensitivity to aggressive discounting.

Category Performance: Technology leads in total sales value, while Furniture drives volume. Office Supplies provide stable, moderate revenue.

Regional Insights: The West and East regions contribute the highest revenue share. The South region shows growth potential.

Customer Concentration: Top 10 customers account for a disproportionate share of revenue, indicating high customer concentration risk.

Discount Impact: Discounts above 30% consistently result in negative profit. A strong negative correlation exists between Discount and Profit Margin.

Correlation: Sales and Profit are strongly positively correlated. Quantity has a moderate positive correlation with Sales. Discount is negatively correlated with Profit.

4. Important Charts

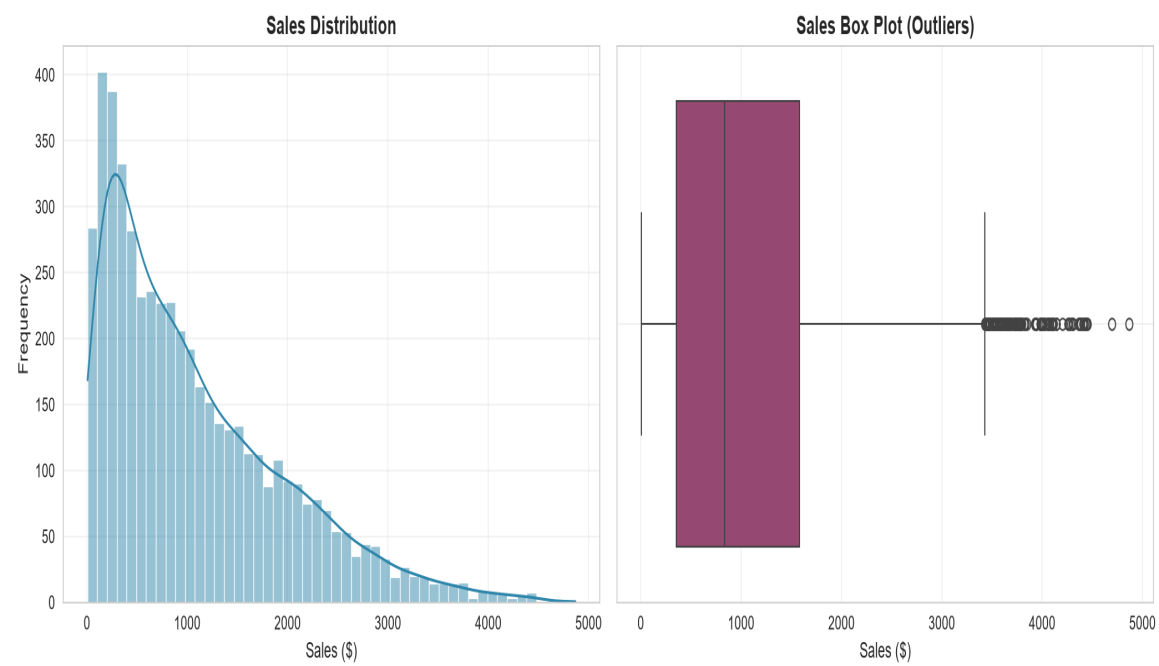


Figure 1: Sales Distribution - Sales Distribution

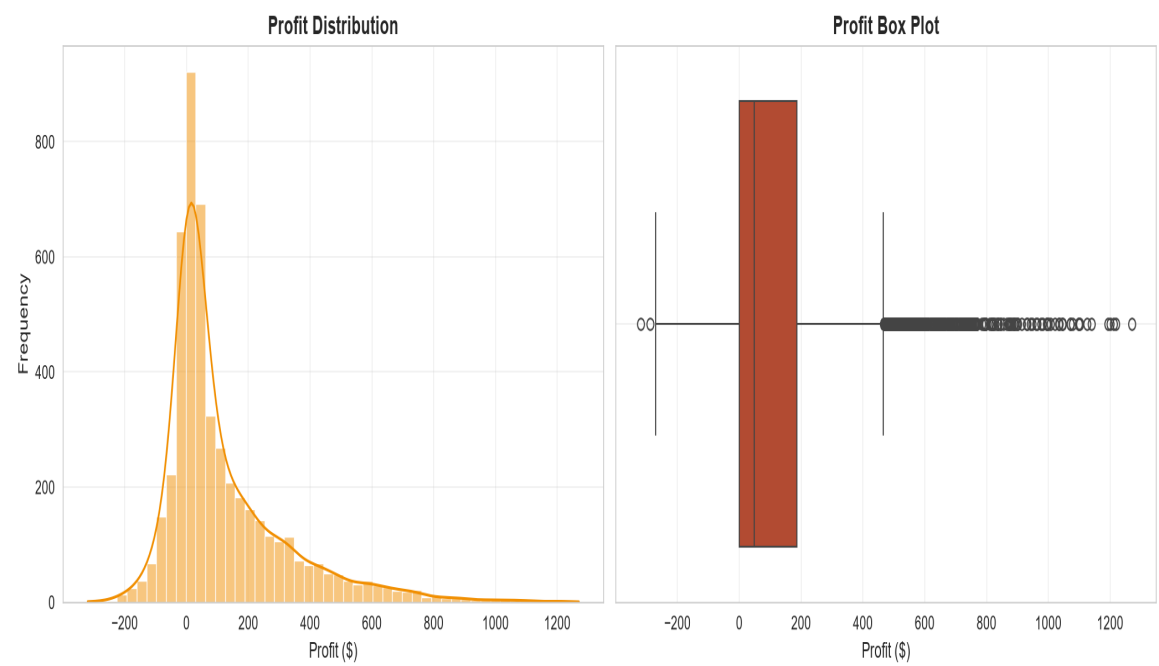


Figure 2: Profit Distribution - Profit Distribution

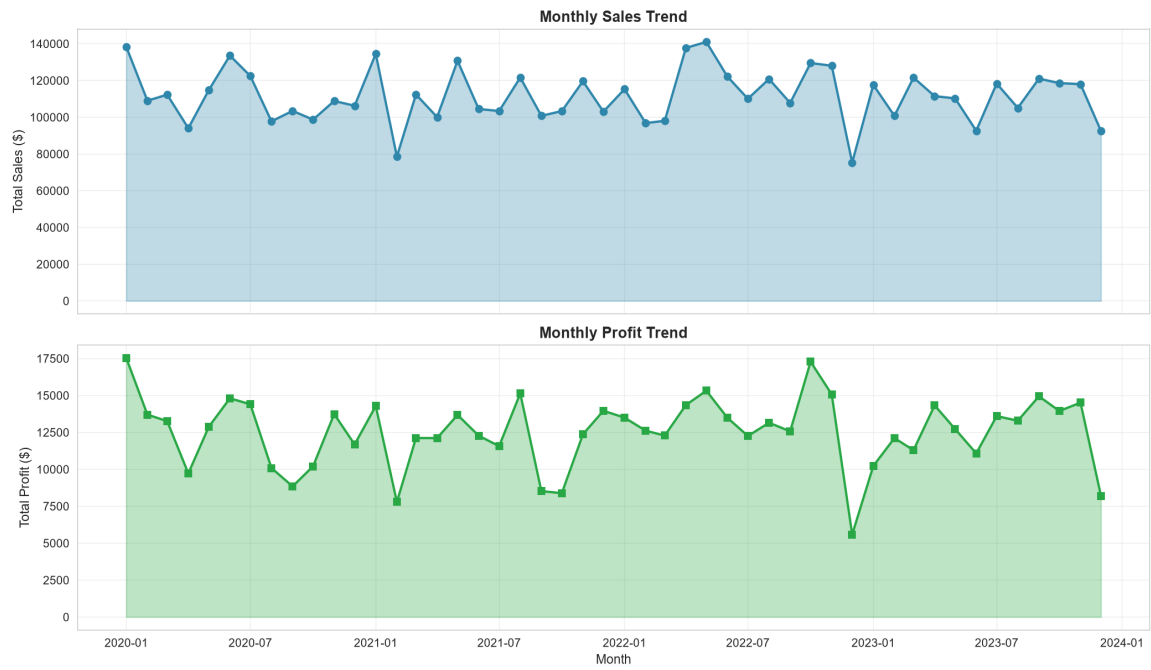


Figure 3: Monthly Sales Trend - Monthly Sales Trend

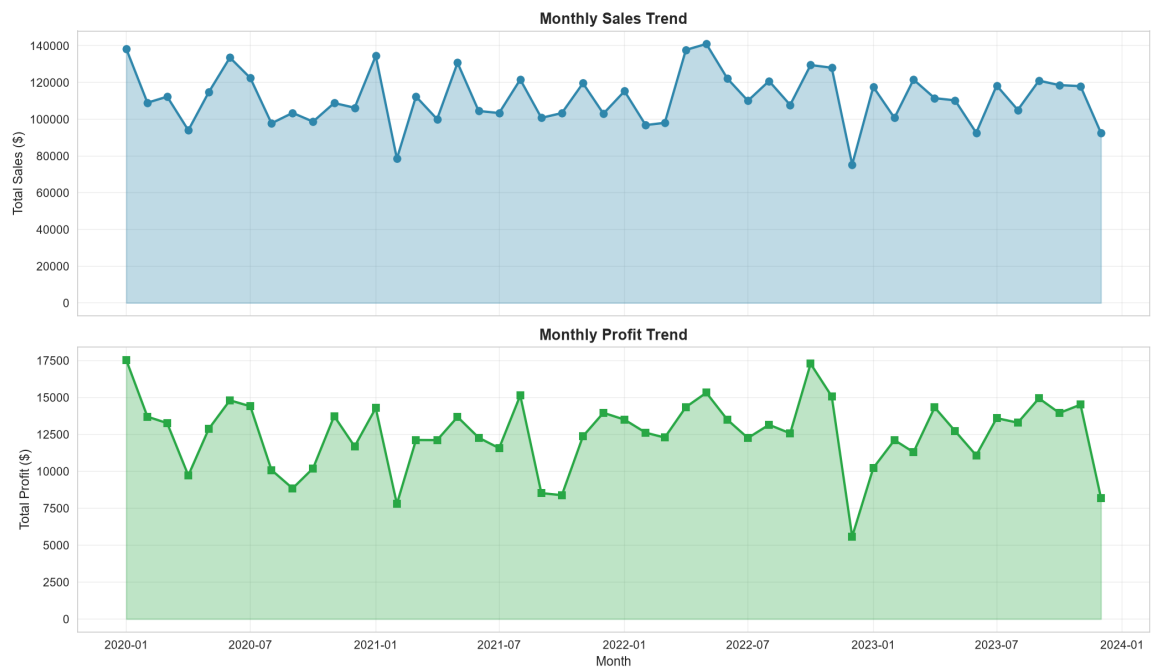


Figure 4: Monthly Profit Trend - Monthly Profit Trend

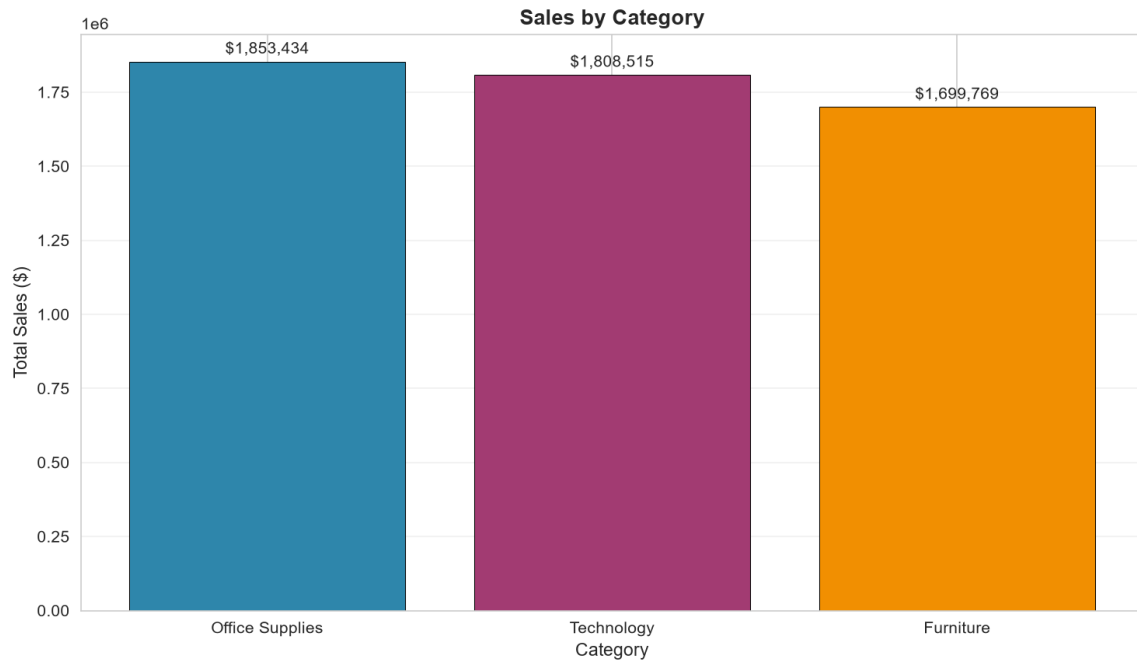


Figure 5: Sales by Category - Sales by Category

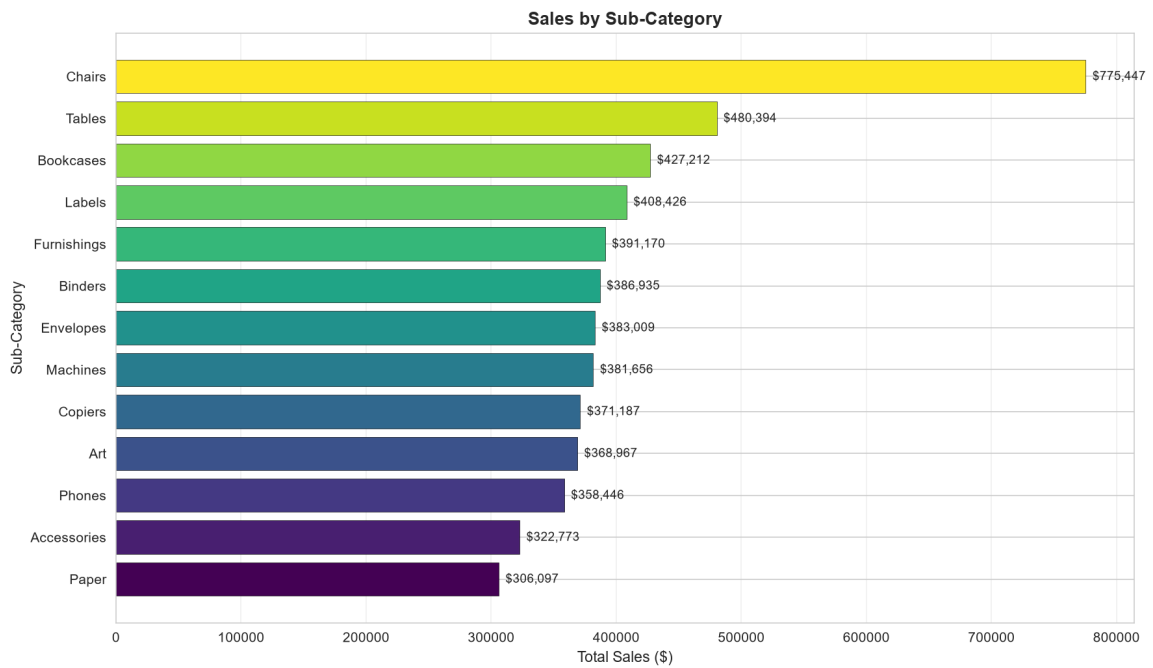


Figure 6: Sales by Sub-Category - Sales by Sub-Category

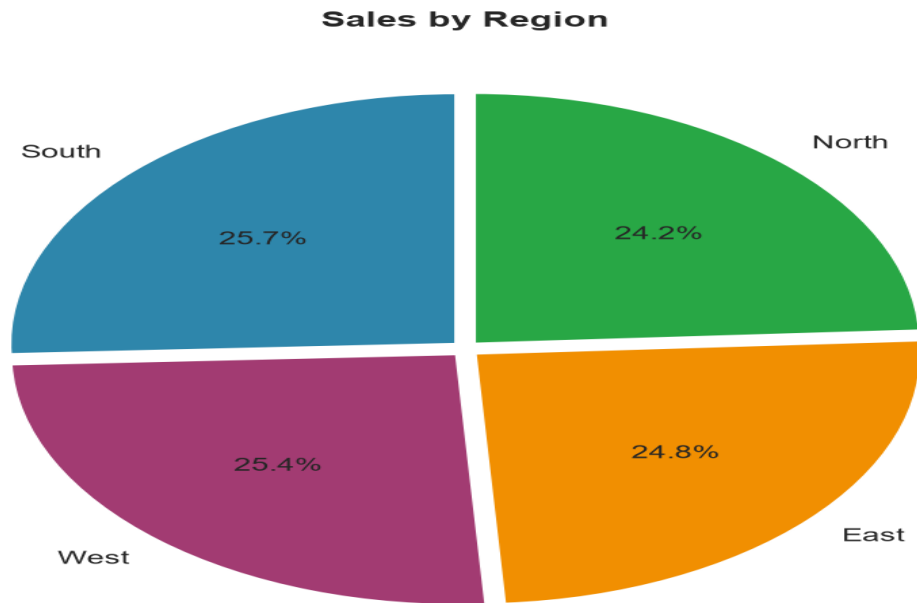


Figure 7: Sales by Region - Sales by Region

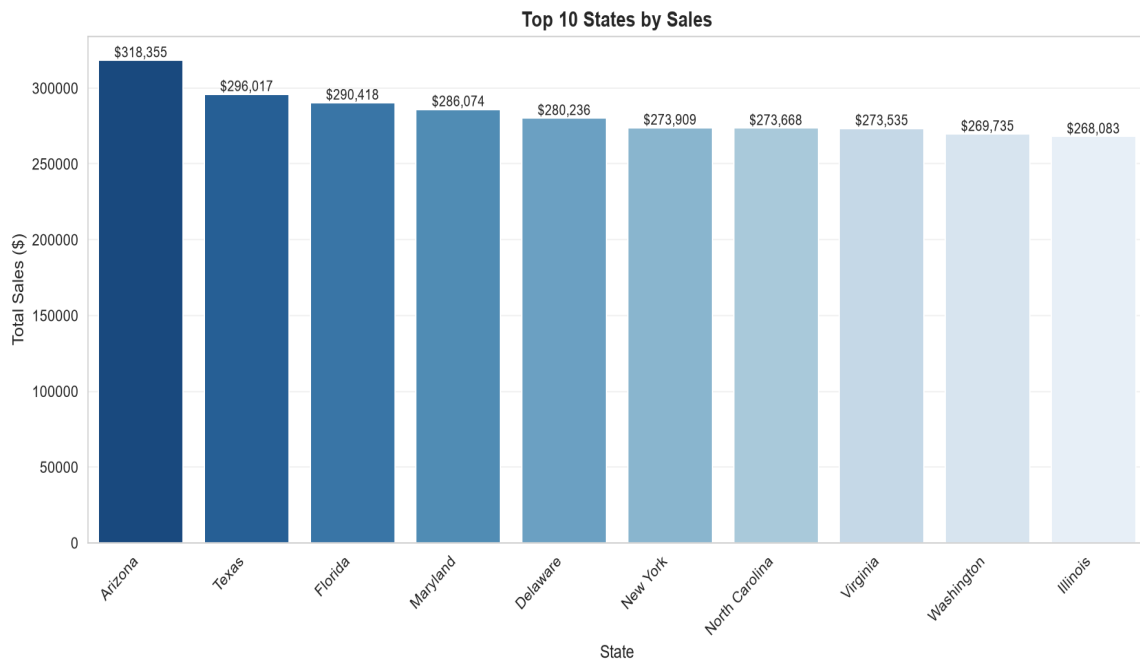


Figure 8: Top 10 States by Sales - Top 10 States by Sales

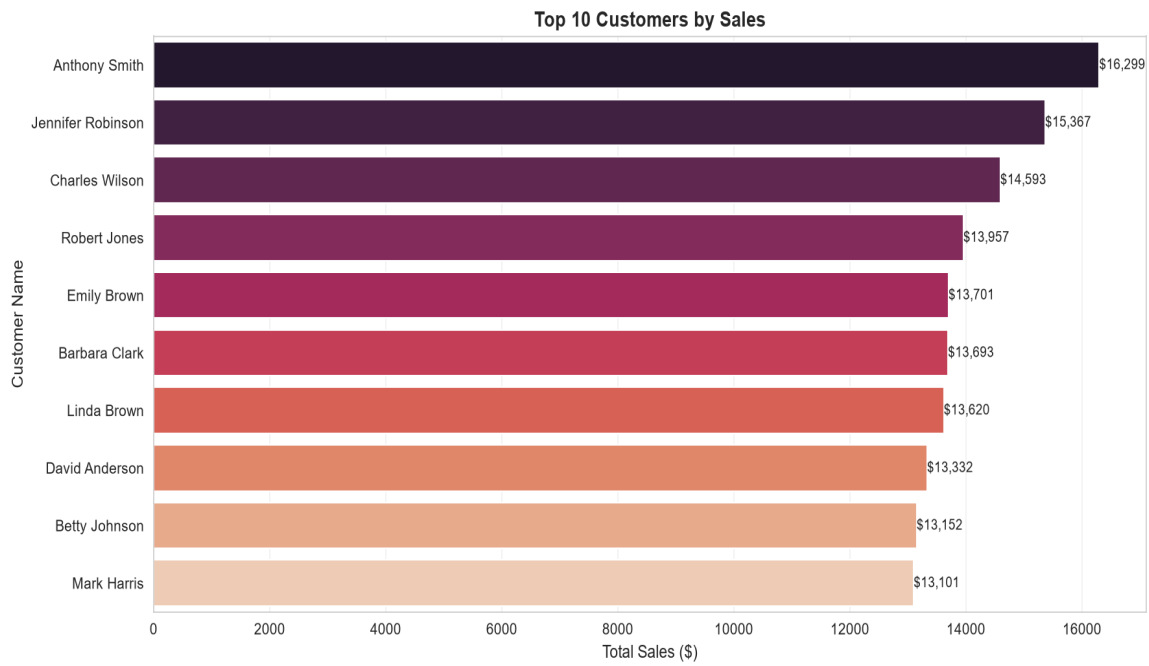


Figure 9: Top 10 Customers - Top 10 Customers



Figure 10: Discount vs Profit - Discount vs Profit

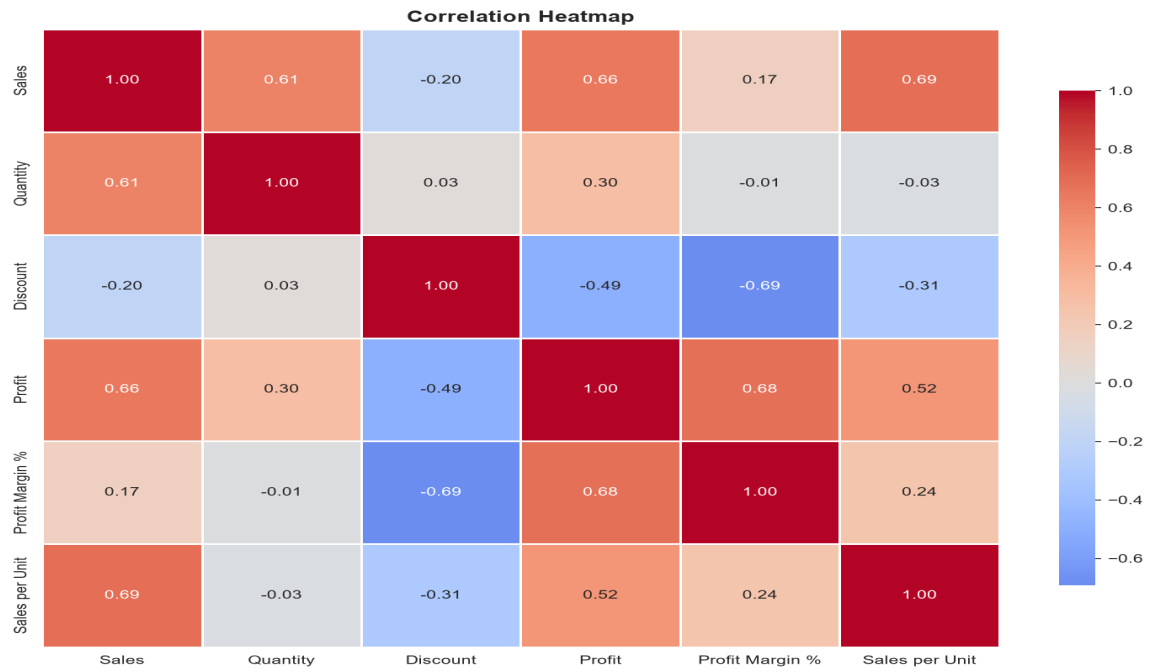


Figure 11: Correlation Heatmap - Correlation Heatmap

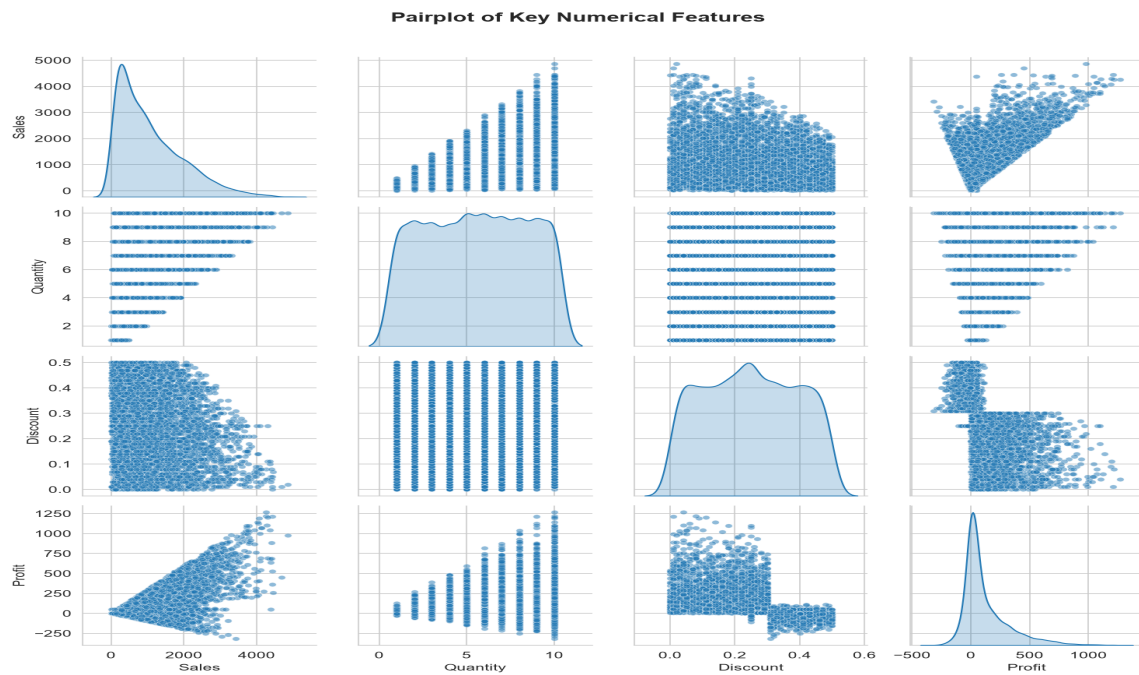


Figure 12: Pairplot - Pairplot of Key Features

5. Recommendations

Discount Policy: Cap discounts at 25% and require manager approval for anything above 15%.

Inventory Planning: Increase stock levels by 20-30% before Q4 and launch promotional campaigns in Q1 to smooth demand.

Category Expansion: Expand Technology SKUs and bundle Furniture with accessories to increase AOV.

Customer Retention: Implement VIP loyalty programs for top 10% customers with exclusive offers.

Regional Focus: Allocate 40% marketing budget to West and East, 35% to South (growth), and 25% to North (maintenance).

Operational Efficiency: Phase out consistently loss-making sub-categories and renegotiate supplier terms.

6. Conclusion

This EDA reveals a retail business with strong seasonal patterns, clear category leaders, and significant opportunities in discount management and customer retention. By focusing on Technology products, controlling discounting, and nurturing high-value customers, the business can achieve sustainable growth and improved profitability. The West and East regions are established markets, while the South presents untapped growth potential.